

Curating Maps And Spatial Data

Empirical Methods — Week 15

Teaching deck draft

- How do researchers turn places, borders, distances, and exposure measures into empirical objects?
- Spatial curation comes before spatial causal inference.
- Geography choice, exposure definitions, crosswalks, and travel costs shape the estimand.
- A map is not evidence until the measurement object is justified and auditable.

Spatial data are economic objects

$$c_{ijm} = \tau_m t_{ijm} + \pi_m d_{ijm} + \kappa_m$$

$$E_i = \sum_j w_{ij} s_j$$

- c_{ijm} : generalized access or commuting cost by mode m .
- s_j : shock, opportunity, amenity, policy, or disamenity at place j .
- w_{ij} : the researcher's exposure technology.
- The unit and weights define the object before any estimator is run.

Spatial objects

Object	Best for	Main risk
Point	addresses, plants, schools, stops	geocoding error, confidentiality, false precision
Polygon	tracts, counties, commuting zones	MAUP, ecological fallacy, arbitrary borders
Raster	pollution, heat, night lights, terrain	resolution, temporal match, aggregation
Network	roads, transit, commute links	mode assumptions, changing schedules, reproducibility
Panel geography	units over time	boundary changes, crosswalk error, support

Geographies and estimands

Research object	Plausible geography	Estimand discipline
Local labor-market shock	commuting zone, county	Is the market integrated by commuting?
Neighborhood exposure	tract, block, residence history	Does timing match the relevant exposure window?
Job access	travel-time isochrone, reachable jobs	Are jobs reachable by the relevant workers?
Policy jurisdiction	city, school zone, county	Does treatment follow the legal boundary?
Market access	network, trade-cost matrix	Are routes and costs measured for the period?

The same raw locations can answer different questions under different geographies.

Exposure and distance decay

$$w_{ij}(h) = \frac{K(d_{ij}/h)}{\sum_{\ell} K(d_{i\ell}/h)}$$

$$V_{ij} = w_j - r_i + A_i - c_{ij} + \varepsilon_{ij}$$

- A small bandwidth makes exposure local; a large bandwidth lets distant places matter.
- Travel costs may be Euclidean, network-based, flow-based, or generalized.
- Net value objects combine wages, rents, amenities, and access costs.
- The curation question is whether the measure matches the mechanism.

Core operations

Operation	What to document
Geocoding	provider, date, match score, failures, manual audit
Projection	input CRS, analysis CRS, distance/area rationale
Spatial join	within, intersects, centroid, nearest, ties, boundary cases
Buffer	radius, mechanism, alternative radii, barriers
Travel time	mode, departure time, routing source, missing routes
Crosswalk	source vintage, target geography, weight type, splits
Aggregation	denominator, weights, baseline year, support
Disclosure control	jitter, masking, suppression, precision loss

Boundary crosswalks

$$X_h = \sum_g a_{gh} X_g$$

- a_{gh} can be area weight, population weight, employment weight, or flow weight.
- Area weights allocate land; population weights allocate residents.
- Employment weights allocate jobs; commuting-flow weights allocate market connections.
- A many-to-many crosswalk should be preserved before any collapsed file is used.

Practical pitfalls

- MAUP: results change with aggregation unit.
- Geocoding error: failures and low-confidence matches are often systematic.
- Temporal mismatch: outcomes, boundaries, addresses, and routes come from different years.
- Endogenous geography choice: radii or units chosen after seeing results.
- Ecological fallacy: area averages interpreted as individual mechanisms.
- Edge effects: access or exposure just outside the study area is omitted.
- Support problems: some places have no comparable observations or valid routes.

Recommended workflow

- 1 State the mechanism before choosing the map layer.
- 2 Define unit of observation, exposure, and variation separately.
- 3 Preserve raw spatial identifiers and source files.
- 4 Validate geocoding, CRS, joins, crosswalks, and routes.
- 5 Build the primary object and one plausible alternative.
- 6 Audit edge cases, missing routes, split units, and out-of-support places.
- 7 Document confidentiality masking and measurement consequences.
- 8 Make the workflow rerunnable without manual GIS clicks.

- Autor, Dorn, Hanson: commuting zones define local labor-market exposure.
- Chetty, Hendren, Katz: tract-level childhood exposure and timing matter.
- Monte, Redding, Rossi-Hansberg: commuting and migration connect local markets.
- Miller: job suburbanization makes access depend on worker and job locations.
- Donaldson, Hornbeck: market access depends on networks and trade costs.

The lesson is not “use GIS.” The lesson is: define the economic object.

Reproduce

- synthetic tracts
- commuting zones
- industry shocks
- baseline weights
- exposure object

Diagnose

- county vs CZ
- crosswalk weights
- geocoding jitter
- edge cases
- estimand memo

Transfer

- job centers
- buffers
- distance decay
- travel time
- access memo

Where curation stops

- Curation defines places, assignments, exposure, distance, and access.
- Curation can diagnose measurement sensitivity and support.
- It does not prove exogeneity, solve sorting, or remove spillovers.
- Spatial causal inference begins when the researcher claims a counterfactual effect.
- Lecture 16 takes up spillovers, interference, border designs, and spatial dependence.

Takeaways

- Geography is part of the estimand.
- Exposure weights are economic assumptions.
- Crosswalks, joins, buffers, and travel times need validation and documentation.
- Sensitivity to plausible geographies is a measurement diagnostic.
- Good spatial work is reproducible, mechanism-driven, and clear about confidentiality.
- Strong spatial curation makes the next lecture's causal designs possible.